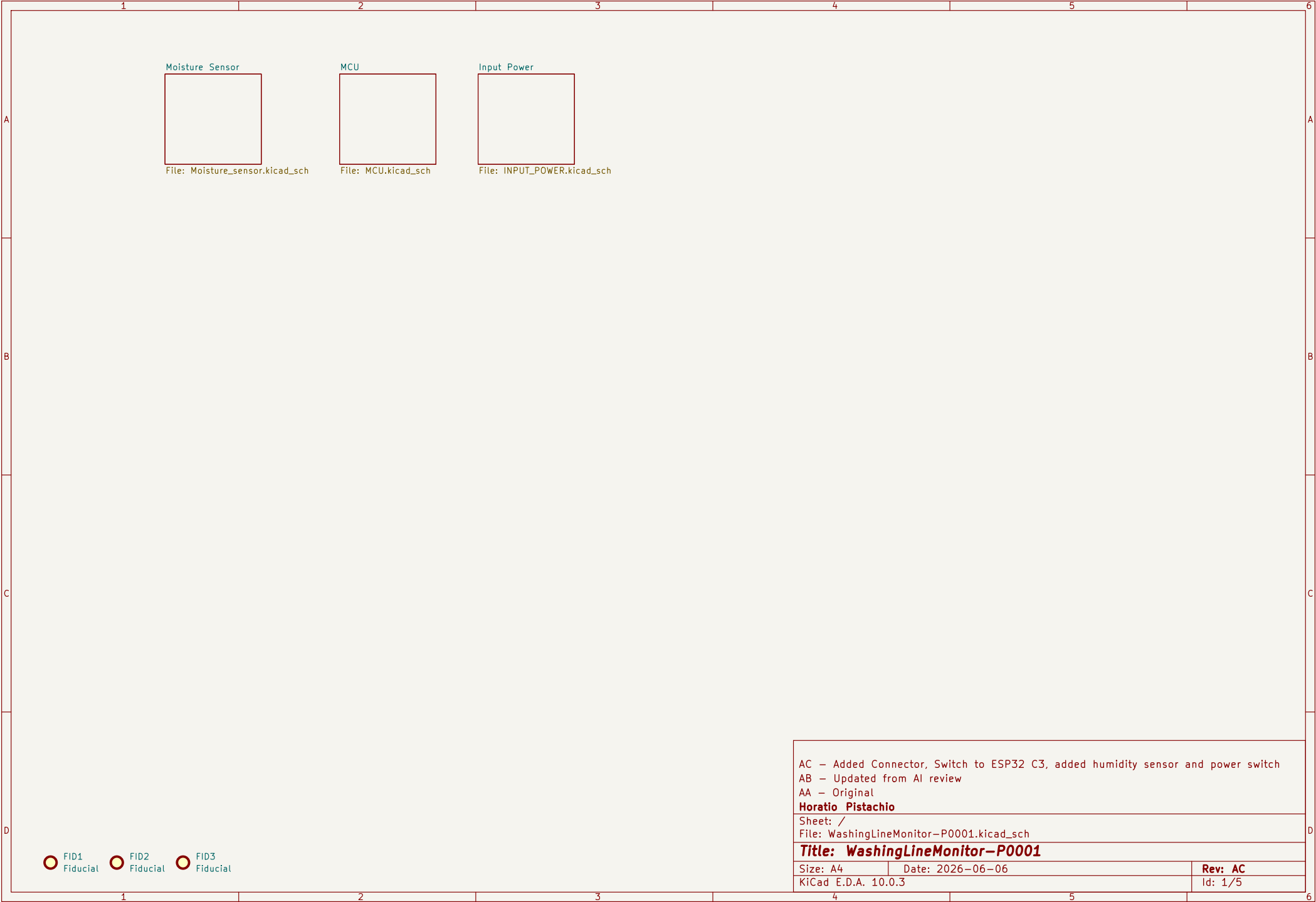
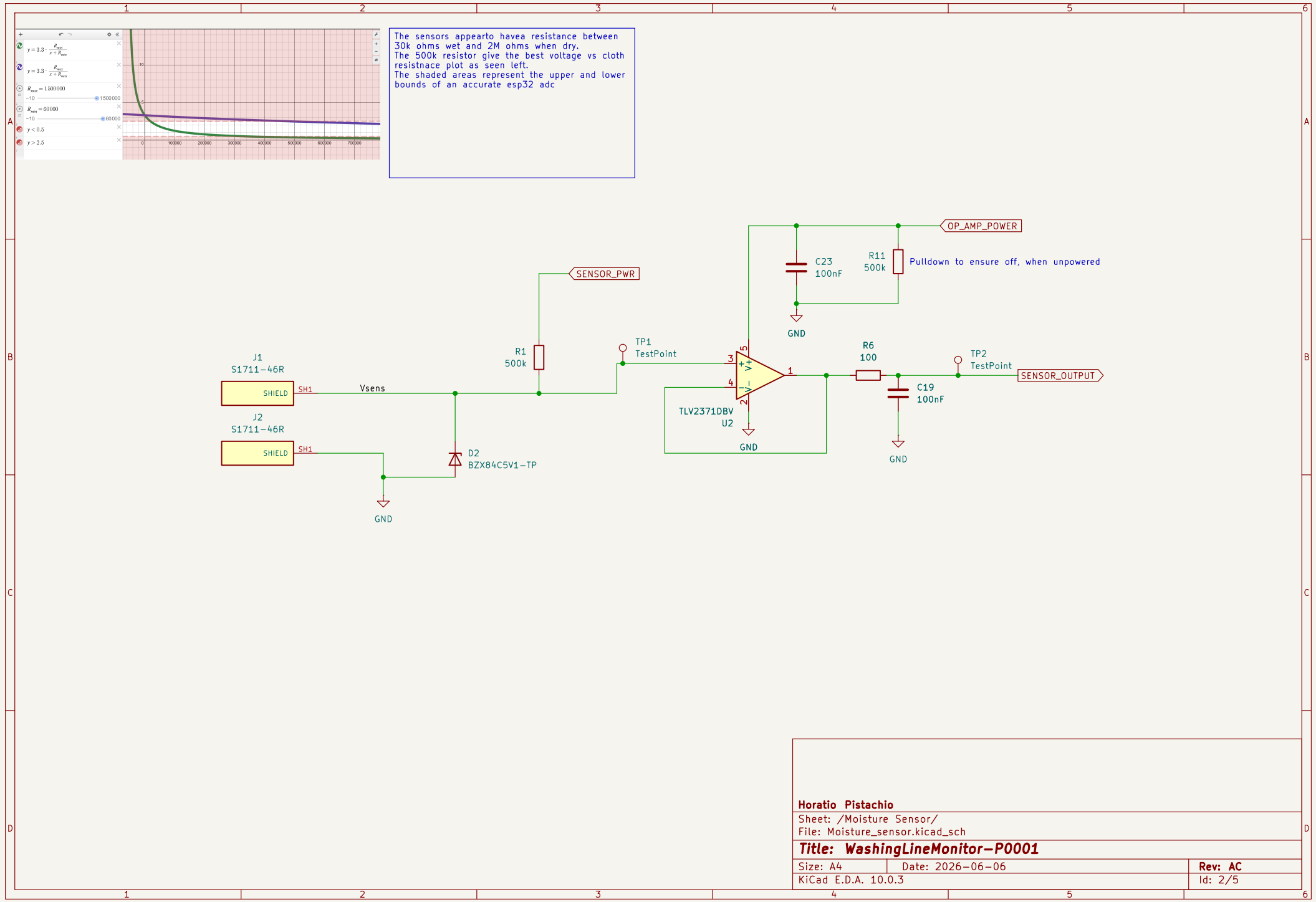
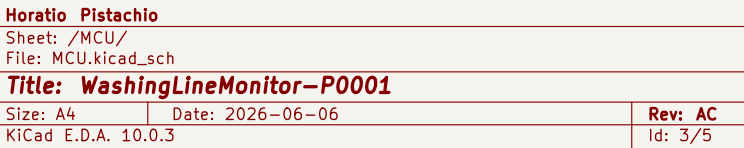
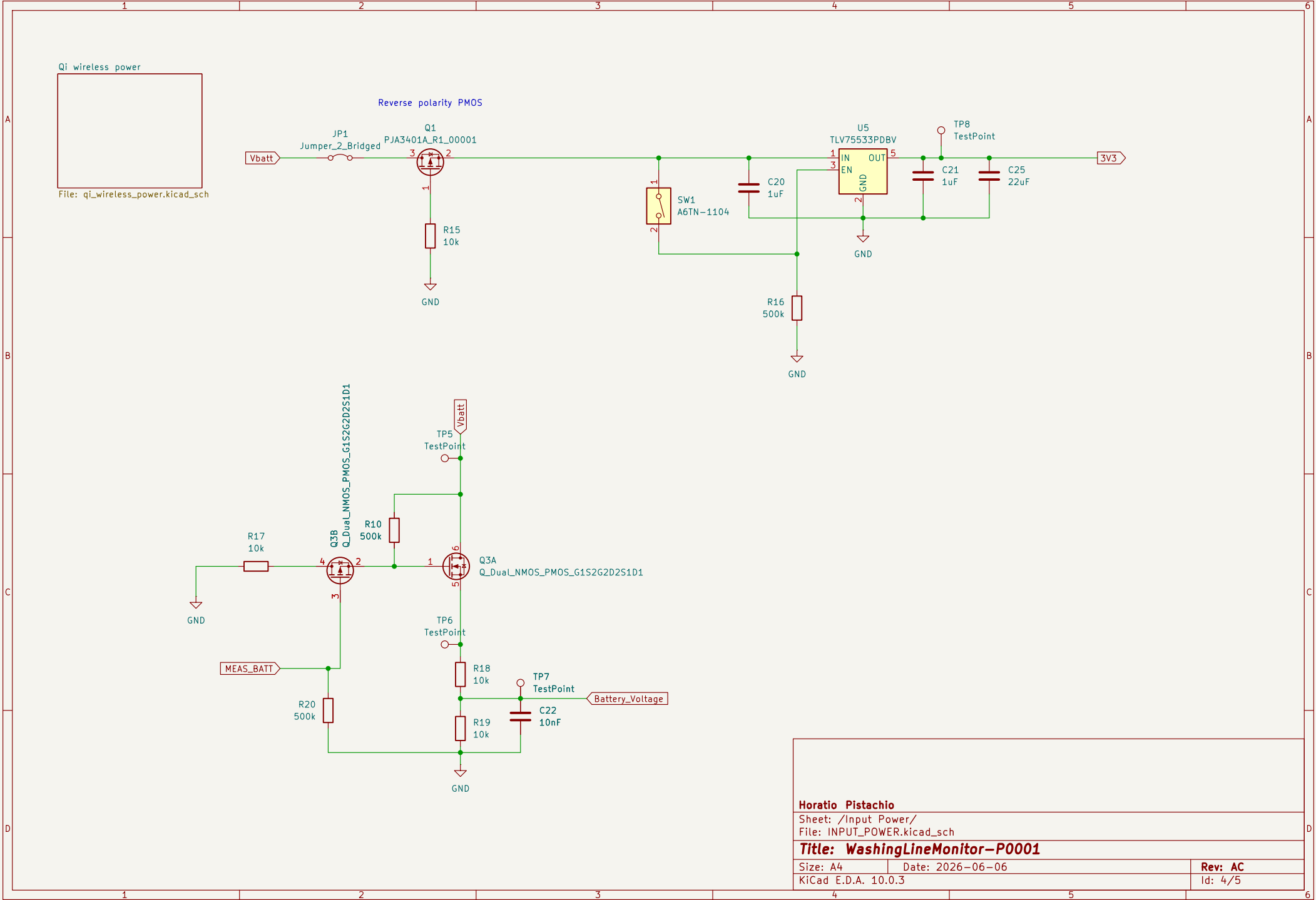




The sensors appear to have resistance between 30k ohms wet and 2M ohms when dry. The 500k resistor gives the best voltage vs cloth resistance plot as seen left. The shaded areas represent the upper and lower bounds of an accurate esp32 adc





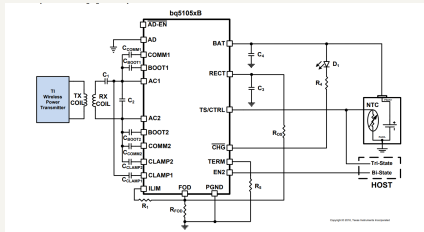


From BQ51050BRHL datasheet application example
Assuming L's (inductance in setup) and Ls (free space inductance)
are the same at 100kHz
L's = 13 uH (WR483245)
R = 0.2 ohm (WR483245)
fs is 100 kHz +5/-10%
fd is 1 MHz ±10%
hence:
C1 = 19uF ~ 10uF + 4.7uF + 4.7uF
C2 = 1.95 nF ~ 1nF + 1nF
Q = 408 (>77 therefore good)

$$C_1 = \frac{1}{(2\pi \times f_s)^2 \times L_s}$$

$$C_2 = \left((f_D \times 2\pi)^2 \times L_s - \frac{1}{C_1} \right)^{-1}$$

$$Q = \frac{2\pi \times f_D \times L_s}{R}$$



Battery charging
Kilim = 314 Aohm (BQ51050)
Ibulk = 500mA (0.5c of a 1000mAh battery)
%Ibulk = 10 (cut off once we are charging at 10% of max)
Rfod = 200ohm (BQ51050)
Kterm = 240 Ohm/%

R1 = 428 ohm ~ 430 ohm
Rterm = 2.4kohm

Rfod is tuneable and recommended to start at 20k and change if needed

8.3.4.3 Battery Charge Current Setting Calculations

8.3.4.3.1 R_{ILIM} Calculations

The bq51050x includes a means of providing hardware overcurrent protection by means of an analog current regulation loop. The hardware current limit provides an extra level of safety by clamping the maximum allowable output current (for example, a current compliance). The calculation for the total R_{ILIM} resistance is as follows:

$$R_{ILIM} = R_1 + R_{FOD} \quad I_{BULK} = \frac{K_{ILIM}}{R_{ILIM}} \quad (1)$$

Where I_{BULK} is the programmed battery charge current during fast charge mode. When referring to the application diagram shown in Figure 32, R_{ILIM} is the sum of R_{FOD} and R₁ (the total resistance from the ILIM pin to PGND).

8.3.4.3.2 Termination Calculations

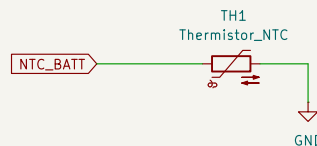
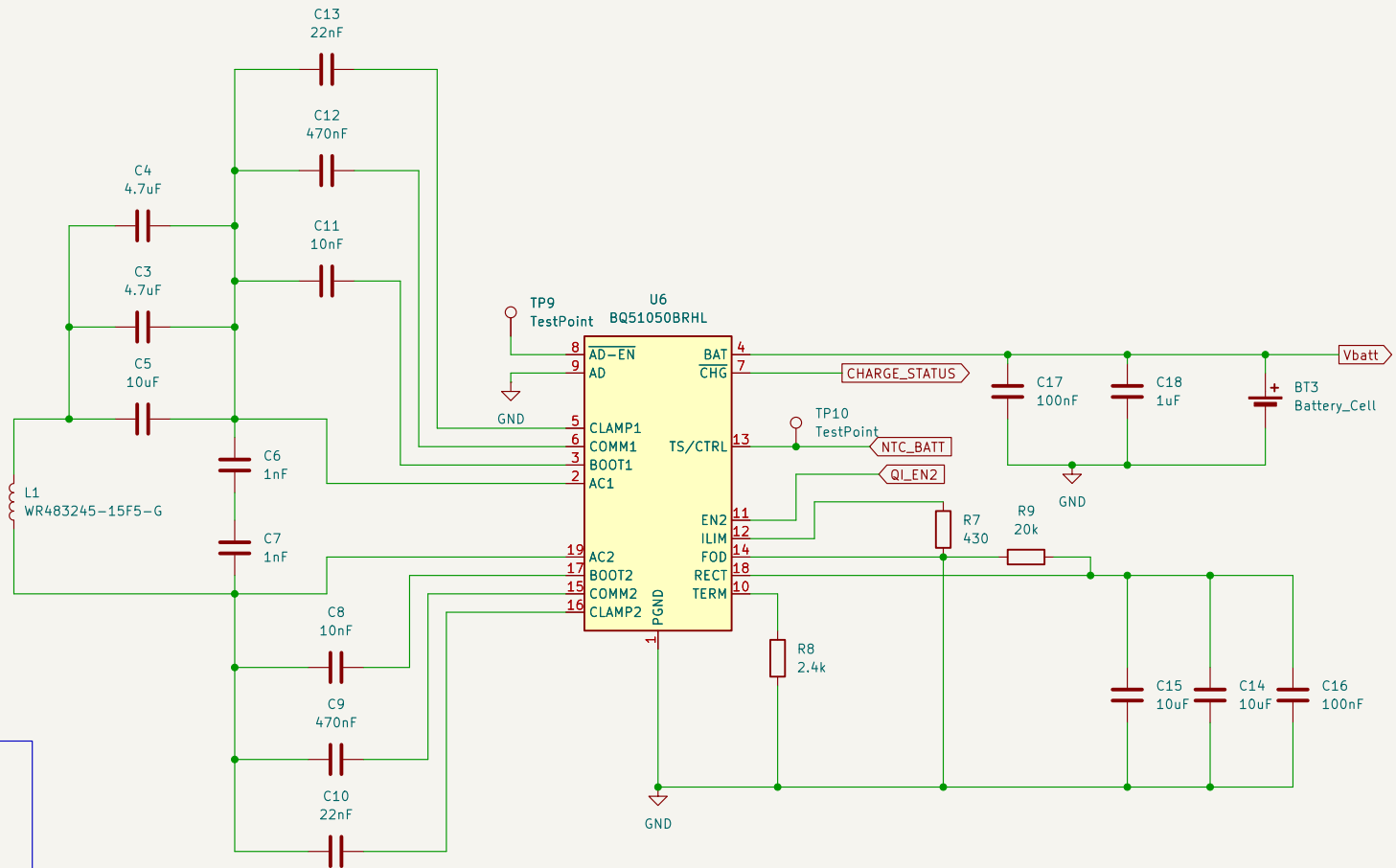
The bq51050x includes a programmable upper termination threshold. The upper termination threshold is calculated using Equation 2:

$$R_{TERM} = K_{TERM} \times \%I_{BULK} \quad \%I_{BULK} = \frac{R_{TERM}}{K_{TERM}} \quad (2)$$

The K_{TERM} constant is specified in Electrical Characteristics as 240 0/1%. The upper termination threshold is set as a percentage of the charge current setting (I_{BULK}).

For example, if R_{ILIM} is set to 314 Ω, I_{BULK} will be 1 A (314 ~ 314). If the upper termination threshold is desired to be 100 mA, this would be 10% of I_{BULK}. The K_{TERM} resistor would then equal 2.4 kΩ (240 × 10).

Termination can be disabled by floating the TERM pin. If the TERM pin is grounded the termination function is effectively disabled. However, due to offsets of internal comparators, termination may occur at low battery currents.



Horatio Pistachio

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